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$$f(x)g(x) \in \mathbb{F}_{<2n}[x]$$
$$fg = \sum_{i+j<2n-2} a_i b_j x^{i+j}$$

Complexity: $O(n^2)$

Suppose $\omega \in \mathbb{F}$ is an nth root of unity.

Recall: if $\mathbb{F} = \mathbb{F}_{p^k}$ then $\exists N : \mathbb{F}_{p^N}$ contains all nth roots of unity.

$$\mathrm{DFT}_\omega : \mathbb{F}^n \to \mathbb{F}^n$$
$$\mathrm{DFT}_\omega(f) = (f(\omega^0), f(\omega^1), ..., f(\omega^{n-1}))$$

$$V_\omega = \begin{pmatrix} 1 & 1 & 1 & ... & 1 \\ 1 & \omega^1 & \omega^2 & ... & \omega^{n-1} \\ 1 & \omega^2 & \omega^4 & ... & \omega^{2(n-1)} \\ \vdots & & & & \\ 1 & \omega^{n-1} & \omega^{2(n-1)} & ... & \omega^{(n-1)^2} \end{pmatrix}$$

$$\mathrm{DFT}_\omega(f) = V_\omega \cdot f^T$$

since vandermonde multiplication is simply evaluation of a polynomial.

Lemma: $V_\omega^{-1} = \frac{1}{n} V_{\omega^{-1}}$

Use $1 + \omega + \cdots + \omega^{n-1}$ and compute $V_\omega V_{\omega^{-1}}$

Corollary: DFT_ω is invertible.

Definitions

1. Convolution $f * g = fg \mod (x^n - 1)$
2. Pointwise product

$$(a_0, ..., a_{n-1}) \cdot (b_0, ..., b_{n-1}) = (a_0 b_0, ..., a_{n-1} b_{n-1}) \in \mathbb{F}^n \to \mathbb{F}_{<n}[x]$$

Theorem: $\mathbf{DFT}_\omega(f * g) = \mathbf{DFT}_\omega(f) \cdot \mathbf{DFT}_\omega(g)$

$$fg = q'(x^n - 1) + f * g$$
$$\Rightarrow f * g = fg + q(x^n - 1)$$
$$\deg fg \leq 2n - 2$$

$$\begin{aligned}(f * g)(\omega^i) &= f(\omega^i)g(\omega^i) + q(\omega^i)(\omega^{in} - 1) \\ &= f(\omega^i)g(\omega^i)\end{aligned}$$

Result

$$f, g \in \mathbb{F}_{<n/2}[x]$$
$$fg = f * g$$
$$\mathrm{DFT}_\omega(f * g) = \mathrm{DFT}_\omega(f) \cdot \mathrm{DFT}_\omega(g)$$
$$fg = \frac{1}{n}\mathrm{DFT}_{\omega^{-1}}(\mathrm{DFT}_\omega(f) \cdot \mathrm{DFT}_\omega(g))$$

Finite Field Extension Containing Nth Roots of Unity

$$\mu_N = \langle\omega\rangle, |\mathbb{F}_{p^N}^\times| = p^N - 1$$
$$\mathrm{ord}(\omega) = n | p^N - 1$$

but $\mathbb{F}_{p^N}^\times$ is cyclic.

For all $d | p^N - 1$, there exists $x \in \mathbb{F}_{p^N}^\times$ with $\mathrm{ord}(x) = d$.

Finding $n | p^N - 1$ is sufficient for $\omega \in \mathbb{F}_{p^N}$

$$n | p^N - 1 \Leftrightarrow \mathrm{ord}(p) = (\mathbb{Z}/n\mathbb{Z})^\times$$

FFT Algorithm Recursive Compute

We recurse to a depth of $\log n$. Since each recursion uses ω^i, then in the final step $\omega^i = 1$, and we simply return f^T.

We only need to prove a single step of the algorithm produces the desired result, and then the correctness is inductively proven.

$$\begin{aligned}f(X) &= a_0 + a_1 X + a_2 X^2 + \cdots + a_{n-1} X^{n-1} \\ &= g(X) + X^{n/2} h(X)\end{aligned}$$

Algorithm

Algorithm 1 Discrete Fourier Transform

function DFT($n = 2^d, f(X)$)
  if $n = 1$ **then**
    return $f(X)$
  end if
  $f(X) = g(X) + X^{n/2}h(X)$ ▷ Write $f(X)$ as the sum of two polynomials with equal degree
  Let $\mathbf{g}, \mathbf{h}$ be the vector representations of $g(X), h(X)$

  $\mathbf{r} = \mathbf{g} + \mathbf{h}$
  $\mathbf{s} = (\mathbf{g} - \mathbf{h}) \cdot (\omega^0, ..., \omega^{n/2-1})$
  Let $r(X), s(X)$ be the polynomials represented by the vectors $\mathbf{r}, \mathbf{s}$

  Compute $(r(\omega^0), ..., r(\omega^{n/2})) = \text{DFT}_{\omega^2}(n/2, r(X))$
  Compute $(s(\omega^0), ..., s(\omega^{n/2})) = \text{DFT}_{\omega^2}(n/2, s(X))$

  return $(r(\omega^0), s(\omega^0), r(\omega^2), s(\omega^2), ..., r(\omega^{n/2}), s(\omega^{n/2}))$
end function

Even Values

$$r(X) = g(X) + h(X)$$

$$\begin{aligned} f(\omega^{2i}) &= g(\omega^{2i}) + (\omega^{2i})^{n/2}h(\omega^{2i}) \\ &= g(\omega^{2i}) + h(\omega^{2i}) \\ &= (g + h)(\omega^{2i}) \end{aligned}$$

So then we can now compute $DFT_\omega(f)_{k=2i} = DFT_{\omega^2}(r)$ for the even powers of $f(\omega^{2i})$.

Odd Values

For odd values $k = 2i + 1$

$$s(X) = (g(X) - h(X)) \cdot (\omega^0, ..., \omega^{n/2-1})$$

$$\begin{aligned} f(X) &= a_0 + a_1X + a_2X^2 + \cdots + a_{n-1}X^{n-1} \\ &= g(X) + X^{n/2}h(X) \\ f(\omega^{2i+1}) &= g(\omega^{2i+1}) + (\omega^{2i+1})^{n/2}h(\omega^{2i+1}) \end{aligned}$$

But observe that for any nth root of unity $\omega^n = 1$ and $\omega^{n/2} = -1$

$$(\omega^{2i+1})^{n/2} = \omega^{in}\omega^{n/2} = \omega^{n/2} = -1$$

$$\begin{aligned} \Rightarrow f(\omega^{2i+1}) &= g(\omega^{2i+1}) - h(\omega^{2i+1}) \\ &= (g - h)(\omega^{2i+1}) \end{aligned}$$

Let $\mathbf{s} = (\mathbf{g} - \mathbf{h}) \cdot (\omega^0, ..., \omega^{n/2-1})$ be the representation for $s(X)$. Then we can see that $s(\omega^{2i}) = (g - h)(\omega^{2i+1})$ as desired.

So then we can now compute $DFT_\omega(f)_{k=2i+1} = DFT_{\omega^2}(s)$ for the odd powers of $f(\omega^{2i+1})$.

Example

Let $n = 8$

$$\begin{aligned} f(X) &= (a_0 + a_1X + a_2X^2 + a_3X^3) + (a_4X^4 + a_5X^5 + a_6X^6 + a_7X^7) \\ &= (a_0 + a_1X + a_2X^2 + a_3X^3) + X^4(a_4 + a_5X + a_6X^2 + a_7X^3) \\ &= g(X) + X^{n/2}h(X) \\ g(X) &= a_0 + a_1X + a_2X^2 + a_3X^3 \\ h(X) &= a_4 + a_5X + a_6X^2 + a_7X^3 \end{aligned}$$

Now vectorize $g(X), h(X)$

$$\begin{aligned} \mathbf{g} &= (a_0, a_1, a_2, a_3) \\ \mathbf{h} &= (a_4, a_5, a_6, a_7) \end{aligned}$$

Compute reduced polynomials in vector form

$$\begin{aligned} \mathbf{r} &= \mathbf{g} + \mathbf{h} \\ &= (a_0 + a_4, a_1 + a_5, a_2 + a_6, a_3 + a_7) \\ \mathbf{s} &= (\mathbf{g} - \mathbf{h}) \cdot (1, \omega, \omega^2, \omega^3) \\ &= (a_0 - a_4, a_1 - a_5, a_2 - a_6, a_3 - a_7) \cdot (1, \omega, \omega^2, \omega^3) \\ &= (a_0 - a_4, \omega(a_1 - a_5), \omega^2(a_2 - a_6), \omega^3(a_3 - a_7)) \end{aligned}$$

Convert them to polynomials from the vectors. We also expand them out below for completeness.

$$\begin{aligned} r(X) &= r_0 + r_1X + r_2X^2 + r_3X^3 \\ &= (a_0 + a_4) + (a_1 + a_5)X + (a_2 + a_6)X^2 + (a_3 + a_7)X^3 \\ s(X) &= s_0 + s_1X + s_2X^2 + s_3X^3 \\ &= (a_0 - a_4) + \omega(a_1 - a_5)X + \omega^2(a_2 - a_6)X^2 + \omega^3(a_3 - a_7)X^3 \end{aligned}$$

Compute

$$\mathrm{DFT}_{\omega^2}(4, r(X)), \mathrm{DFT}_{\omega^2}(4, s(X))$$

The values returned will be

$$(r(1), s(1), r(\omega^2), s(\omega^2), r(\omega^4), s(\omega^4), r(\omega^6), s(\omega^6)) = (f(1), f(\omega), f(\omega^2), f(\omega^3), f(\omega^4), f(\omega^5), f(\omega^6), f(\omega^7))$$

Which is the output we return.

Comparing Evaluations for $f(X)$ and $r(X), s(X)$

We can see the evaluations are correct by substituting in ω^i.

We expect that $s(X)$ on the domain $(1, \omega^2, \omega^4, \omega^6)$ produces the values $(f(1), f(\omega^2), f(\omega^4), f(\omega^6))$, while $r(X)$ on the same domain produces $(f(\omega), f(\omega^3), f(\omega^5), f(\omega^7))$.

Even Values

Let $k = 2i$, be an even number. Then note that k is a multiple of 2, so $4k$ is a multiple of $n \Rightarrow \omega^{4k} = 1$,

$$\begin{aligned} r(X) &= (a_0 + a_4) + (a_1 + a_5)X + (a_2 + a_6)X^2 + (a_3 + a_7)X^3 \\ r(\omega^{2i}) &= (a_0 + a_4) + (a_1 + a_5)\omega^{2i} + (a_2 + a_6)\omega^{4i} + (a_3 + a_7)\omega^{6i} \\ f(\omega^k) &= (a_0 + a_1\omega^k + a_2\omega^{2k} + a_3\omega^{3k}) + \omega^{4k}(a_4 + a_5\omega^k + a_6\omega^{2k} + a_7\omega^{3k}) \\ &= (a_0 + a_1\omega^k + a_2\omega^{2k} + a_3\omega^{3k}) + (a_4 + a_5\omega^k + a_6\omega^{2k} + a_7\omega^{3k}) \\ &= (a_0 + a_4) + (a_1 + a_5)\omega^k + (a_2 + a_6)\omega^{2k} + (a_3 + a_7)\omega^{3k} \\ &= f(\omega^{2i}) \\ &= (a_0 + a_4) + (a_1 + a_5)\omega^{2i} + (a_2 + a_6)\omega^{4i} + (a_3 + a_7)\omega^{6i} \\ &= r(\omega^{2i}) \end{aligned}$$

Odd Values

For $k = 2i + 1$ odd, we have a similar relation where $4k = 8i + 4$, so $\omega^{4k} = \omega^4$. But observe that $\omega^4 = -1$.

$$
\begin{aligned}
s(X) &= (a_0 - a_4) + \omega(a_1 - a_5)X + \omega^2(a_2 - a_6)X^2 + \omega^3(a_3 - a_7)X^3 \\
s(\omega^{2i}) &= (a_0 - a_4) + (a_1 - a_5)\omega^{2i+1} + (a_2 - a_6)\omega^{4i+2} + (a_3 - a_7)\omega^{6i+3} \\
f(\omega^k) &= (a_0 + a_1\omega^k + a_2\omega^{2k} + a_3\omega^{3k}) + \omega^{4k}(a_4 + a_5\omega^k + a_6\omega^{2k} + a_7\omega^{3k}) \\
&= (a_0 + a_1\omega^k + a_2\omega^{2k} + a_3\omega^{3k}) - (a_4 + a_5\omega^k + a_6\omega^{2k} + a_7\omega^{3k}) \\
&= f(\omega^{2i+1}) \\
&= (a_0 + a_1\omega^{2i+1} + a_2\omega^{4i+2} + a_3\omega^{6i+3}) - (a_4 + a_5\omega^{2i+1} + a_6\omega^{4i+2} + a_7\omega^{6i+3}) \\
&= (a_0 - a_4) + (a_1 - a_5)\omega^{2i+1} + (a_2 - a_6)\omega^{4i+2} + (a_3 - a_7)\omega^{6i+3} \\
&= s(\omega^{2i})
\end{aligned}
$$